(a) Tactile description

Paired fingertip signals · last T frames

P2L
Left tactile clip

P2R
Right tactile clip

t − T + 1 … t

Shared spatial CNN

Temporal convolution

Discrete tactile text

area · Fx · Fy · Fz · rotation

Reference: left-finger sensor frame

(b) Shared reasoning and action model

Updated observations

Tactile semantics

P1F
Front view

P1W
Wrist view

State + task
qpos · instruction

Task-specific context
Touch · motion history
current plan · recovery memory

Visual observations · SigLIP

Shared $\pi_{0.5}$ / PaliGemma backbone
Task-conditioned calls · shared parameters

Shared language decoding

Failure diagnosis
Global-frame failure

Recovery planning
Direction + magnitude

plan

Recovery memory
Executed plan → subsequent failure

Action Expert
Flow matching
H30 action chunk

P3
Robot interaction

Actions use vision, state and task / plan

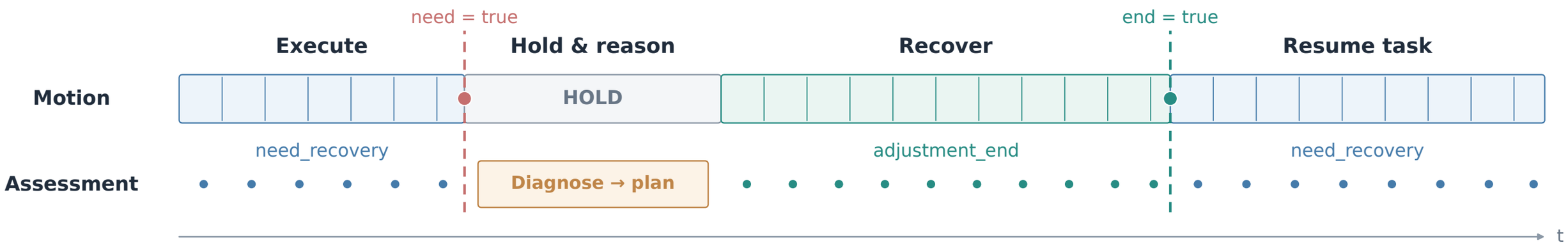
Trigger recovery
need_recovery

End recovery
adjustment_end

Two independent binary heads

(c) Asynchronous execution–recovery loop

Monitor during motion · interrupt within a chunk



Resume, re-evaluate, and recover again if a fault remains.

Solid: information / action flow Dashed: feedback / stage control • Shared-model calls are serialized; monitoring overlaps physical motion.